# Outline No.: 02

**Date:** **Time:** **Week:**

| SL No. | Title | Content | Remark |
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| **1** | **Topic** | Solution of first order & first degree ordinary differential equations by various methods | CO1, PO1 |
| **1.1** | **Pre-requisite Topics** | Basic Calculus Concepts (Differentiation and Integration), Concept of a Differential Equation (Definition, Order and Degree), Types of First-Order ODEs (Linear and Nonlinear ODEs, Homogeneous and Non-homogeneous Equations) | |
| **1.2** | **Learning Outcomes** | Learning outcomes for introduction to differential equations & their formations are:<br>**a. Understanding the Nature of First-Order and First-Degree ODEs**<br>• Recognize the Structure: Identify first-order and first-degree ODEs, understanding that these equations involve only the first derivative and that the highest power of the derivative is 1.<br>• Classify ODEs: Distinguish between linear and nonlinear ODEs, and understand their importance in selecting appropriate solution methods.<br>**b. Applying Specific Methods to Solve First-Order ODEs**<br>• Separation of variables: R-11<br>• Transformation of some equations in the form in which variables are separable: R-15<br>• Homogeneous differential equation: R-18<br>• Equation reducible to homogeneous form :R-22<br>• Exact differential equation: R-26<br>• Linear differential equation: R-42<br>• Equation reducible to linear form: R-48<br>**c. Interpretation of Solutions**<br>• Behavior of Solutions: Understand the qualitative behavior of solutions to differential equations, such as stability, growth, or oscillation, based on initial conditions.<br>• General vs. Particular Solutions: Ability to distinguish between general solutions (involving arbitrary constants) and particular solutions (specific to initial conditions). | |
| **1.3** | **Activities/ Teaching and Learning Strategies (TLS)** | Interactive lectures, Audio/Video, Individual/Group discussion, To do, Feedback, Homework<br><br>(TS1, TS2, TS3, TS4) | |
| **1.4** | **Reading** | Ordinary and Partial differential Equations - M. D Raisinghania; Part- I: Elementary Differential Equations (Ordinary Differential Equations); Chapter-02 (Equations First Order & First Degree); Page 2.1 | |
| **1.5** | **Resources** | Some of resources are:<br>• Ordinary and Partial differential Equations - M. D Raisinghania<br>• Schaum's Outline of Differential Equations - Richard Bronson and Gabriel Costa (Mc Graw-Hill)<br>• Differential Equations - M. L. Khanna<br>• Internet | |

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| **1.6** | **Historical Background** | The historical background of differential equations of first order with first degree is deeply tied to the development of calculus, as well as the broader evolution of mathematics during the 17th, 18th, and 19th centuries.<br>**a. Early Concepts and Foundations (Pre-17th Century)**<br>Though differential equations were not formally recognized in ancient times, early mathematicians had worked with problems that involved rates of change and geometric methods that would eventually lead to the development of differential equations.<br>• Ancient Greeks and Egyptians had mathematical knowledge and practical applications that touched on ideas of change, such as working with concepts of motion, growth, and geometric curves. However, the lack of formalized calculus meant these concepts were more geometric or algebraic rather than expressed as derivatives.<br>• Islamic Golden Age (8th–12th century): Scholars such as Al-Khwarizmi (9th century) made foundational contributions to algebra, which would later be applied in solving equations of different forms, including those that would become differential equations. They advanced the algebraic understanding of linear equations, though not directly connected to differential equations.<br><br>**b. The Birth of Calculus and Differential Equations (17th Century)**<br>The 17th century saw the formal creation of calculus, which provided the tools necessary to express and solve differential equations.<br>• Isaac Newton and Gottfried Wilhelm Leibniz (late 1600s) independently developed the core principles of calculus, including differentiation and integration. This new framework revolutionized mathematics and provided the means to study rates of change—foundational for differential equations. Newton, in particular, used derivatives to describe the laws of motion, setting the stage for later developments in differential equations.<br>For example, Newton's second law of motion can be formulated as a second-order differential equation, but the idea of modeling physical systems in terms of change (rates of change) emerged with his and Leibniz's work.<br>• Differential Equations: In the context of calculus, the first-order, first-degree differential equation refers to the simplest type of differential equation involving a single derivative. Early instances of first-order equations arose from trying to describe the motion of objects or the growth of populations (e.g., the rate of change of position or the spread of an infection).<br><br>**c. Euler and the Development of First-Order Equations (18th Century)**<br>The 18th century saw significant advances in both the theory and methodology of solving differential equations, particularly by Leonhard Euler. | |

- Euler's Contributions: Euler made significant contributions to the formal study and solution of differential equations. He developed general methods to solve first-order differential equations, especially the linear equations of the form:
- dx
  dy +P(x)y=Q(x)
- This type of equation, called a linear differential equation, became a central object of study. Euler also worked with separable equations, a form of first-order differential equations where the variables $x$ and y can be separated and integrated independently.
- Euler's work was influential in the development of applied mathematics, as these equations could now be applied to problems in physics, engineering, and other sciences.
- Joseph-Louis Lagrange: Lagrange also contributed significantly to the development of first-order differential equations, particularly in the context of mechanical systems. He expanded the theory and application of first-order differential equations in his formulation of Lagrangian mechanics.

**d. Formalization and Further Methods (19th Century)**

As the study of differential equations progressed, more systematic and generalized methods for solving first-order differential equations were developed.

- Pierre-Simon Laplace and Augustin-Louis Cauchy made notable contributions to the field in the 19th century. Cauchy, in particular, helped formalize the Cauchy problem, which deals with solving differential equations given initial conditions. This became essential for applying differential equations to real-world problems in physics and engineering.
- Bernoulli’s Equation: The study of more specialized types of first-order equations, such as Bernoulli's equation, was introduced by Jacob Bernoulli in the late 17th and early 18th centuries. These types of equations involve non-linear terms but are still considered first-order and first-degree equations.

**e. Separation of Variables and Modern Techniques (19th Century and Beyond)**

By the 19th century, the method of separation of variables became a key tool for solving first-order differential equations of the form:

dx
dy =g(x)h(y)

- Separation of Variables involves manipulating the equation so that all terms involving y are on one side and all terms involving x are on the other side, allowing the equation to be integrated directly.

This technique became one of the most important tools for solving first-order differential equations and was widely used in areas such as population modeling, heat conduction, and motion.

**f. Numerical Methods and Applications (20th Century to Present)**

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| | | In the 20th century, as more complex problems in physics, biology, and engineering emerged, numerical methods for solving differential equations became crucial. These methods, such as Euler's method and the Runge-Kutta methods, were developed to approximate solutions when an exact analytical solution is difficult or impossible to obtain.<br>• Computational Mathematics: The rise of computers in the 20th century transformed the study and application of first-order differential equations. Numerical solutions allowed for the analysis of more complex systems that could not be solved using traditional analytical methods.<br>• Applications: Differential equations of first-order with first-degree terms have applications in numerous fields:<br>Physics: Modeling systems like fluid flow, electrical circuits, or radioactive decay.<br>Biology: Population models, the spread of diseases, or enzyme reactions.<br>Economics: Modeling economic growth, inflation rates, or investment returns.<br><br>The study of first-order differential equations has continued to be a central aspect of mathematical analysis, playing an essential role in applied sciences and engineering throughout history. | |
| **1.7** | **Applications** | **First-order, first-degree differential equations** are among the simplest and most widely applied types of differential equations in real-world problems. These equations are of the form:<br>$dy/dx=f(x,y)$<br>or in some cases, they can be written as:<br>$dy/dx+P(x)y=Q(x)$<br>These equations describe various phenomena where the rate of change of a quantity is related to that quantity and possibly other factors. Below are several key **applications** of first-order, first-degree differential equations in various fields.<br>a. **Population Dynamics (Biology and Ecology)**<br>One of the most common applications of first-order, first-degree differential equations is in modeling **population growth** and **decay**.<br>**Exponential Growth**: When a population grows at a rate proportional to its size, it can be modeled by the differential equation:<br>$dP/dt=rP$<br>where P(t) is the population at time t, and r is the growth rate. This leads to the **exponential growth equation** $P(t)=P_0e^{rt}$ where $P_0$ is the initial population.<br>**Population Decay**: If the population decays at a rate proportional to its size, the equation becomes:<br>$dP/dt=-rP$<br>This models phenomena like radioactive decay or the decline in the | |

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| | | number of individuals in a species over time.<br><br>**Logistic Growth**: In real-world systems, growth often slows as the population reaches its environment's carrying capacity. A **logistic growth model** is also a first-order differential equation:<br><br>$dP/dt=rP(1-P/K)$<br><br>where K is the carrying capacity of the environment.<br><br>**b. Radioactive Decay (Physics)**<br><br>Radioactive decay is another classic example of a first-order, first-degree differential equation. The rate at which a substance decays is proportional to the amount of substance present at any given time. This is governed by the equation:<br><br>$dN/dt=-\lambda N$<br><br>where:<br><br>N(t)) is the amount of radioactive substance at time t,<br><br>$\lambda$ is the decay constant,<br><br>t is time.<br><br>This equation leads to the solution:<br><br>$N(t)=N_0e-^{\lambda t}$<br><br>where $N_0$ is the initial amount of substance. This exponential decay model is critical in fields like nuclear physics, medicine (e.g., for radiotherapy), and archaeology (e.g., carbon dating).<br><br>**c. Newton's Law of Cooling (Thermodynamics)**<br><br>In thermodynamics, **Newton's Law of Cooling** describes the rate at which the temperature of an object changes relative to its surroundings. The law is expressed as a first-order, first-degree differential equation:<br><br>$dT/dt=-k(T-T_{env})$<br><br>where:<br><br>T(t) is the temperature of the object at time t,<br><br>$T_{env}$ is the constant temperature of the surrounding environment,<br><br>k is a positive constant related to the properties of the object and the environment.<br><br>This equation models the cooling of a hot object or the warming of a cold object over time. The solution is:<br><br>$T(t)= T_{env} +(T_0- T_{env})e^{-kt}$<br><br>where $T_0$ is the initial temperature of the object.<br><br>**d. Electric Circuits (Engineering)**<br><br>In electrical engineering, **first-order linear differential equations** model the behavior of electrical circuits, particularly **RC (Resistor-Capacitor) circuits**. The voltage across a capacitor in an RC circuit is governed by the following equation: | |

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| | | $dV/dt=(1/RC)(V_0-V)$<br>where:<br>$V(t)$ is the voltage across the capacitor at time ttt,<br>$V_0$ is the source voltage,<br>R is the resistance,<br>C is the capacitance.<br>This equation describes how the voltage across the capacitor changes over time when a current is applied, and the solution gives the charging or discharging behavior of the capacitor.<br>**e. Chemical Kinetics (Chemistry)**<br>In chemistry, **reaction rates** often follow first-order differential equations, especially for reactions where the rate is proportional to the concentration of a reactant.<br>For a **first-order reaction**, the rate law is given by:<br>$d[A]/dt=-k[A]$<br>where [A] is the concentration of the reactant at time t and k is the rate constant. The solution to this equation is:<br>$[A](t)=[A]_0e^{-kt}$<br>where $[A]_0$ is the initial concentration of the reactant. This model applies to many chemical reactions, such as radioactive decay, first-order reactions in solution, and enzyme kinetics.<br>**f. Heat Conduction (Engineering and Physics)**<br>The **rate of heat conduction** in a material can also be described by a first-order, first-degree differential equation. According to **Fourier's Law of Heat Conduction**, the rate of change of temperature within a material is proportional to the gradient of the temperature. In the case of a one-dimensional rod, this can be expressed as:<br>$dT/dt=\alpha d^2T/dx^2$<br>where $T(x,t)$ is the temperature at position xxx and time t, and $\alpha$ is the thermal diffusivity constant. However, a simpler case for heat transfer in a material may involve **exponential decay** of temperature over time, which can also be formulated as a first-order equation.<br>**g. Financial Models (Economics)**<br>In economics, differential equations can model the rate of change of financial quantities, such as interest rates or investments.<br>**Continuous Compounding**: If money is invested in an account that earns interest at a rate proportional to the current amount in the account, the amount of money $A(t)$ at time ttt follows the differential equation:<br>$dA/dt=rA$<br>where r is the rate of return. The solution to this equation is:<br>$A(t)=A_0e^{rt}$ | |

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| | | This represents **continuous compound interest**, where $A_0$ is the initial amount of money.<br><br>**h. Simple Harmonic Motion (Physics)**<br><br>In physics, the motion of objects under a restoring force (such as a mass on a spring) is often modeled by a second-order differential equation, but for simplicity, if we consider the velocity or displacement over time, first-order approximations can be made.<br><br>These equations provide essential tools for understanding and solving problems involving rates of change in diverse scientific disciplines. | |
| **1.8** | **Definitions** | Some of useful definitions are:<br>**a.** First order and first degree differential equation<br>R (Raisinghania)-2.1(Page) | |
| **1.9** | **Various Methods** | Various Methods to Solve First Order and First Degree Differential Equations:<br>**a.** Separation of variables R-2.1<br>**b.** Transformation of some equations in the form in which variables are separable R-2.4<br>**c.** Homogeneous differential equation R-2.7<br>**d.** Equation reducible to homogeneous form R-2.11<br>**e.** Exact differential equation R-2.16<br>**f.** Linear differential equation R-2.1<br>**g.** Equation reducible to linear form R-2.1 | |
| **1.10** | **Problems** (Separation of Variables) | Separation of Variables:<br>**a.** Standard form R-2.1<br>$M(x,y,c)dx + N(x,y,c)dy = 0 \text{ or } \frac{dy}{dx} = f(x,y,c)$<br>$\Rightarrow f(x,c)dx + g(y,c)dy = 0$<br>**b.** Working rule to solve an equation in which variables are separable R-2.1<br>**Solve the following differential equations:**<br>**a.** $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$ R-2.1-2.3-1<br>**b.** $y - x\frac{dy}{dx} = a(y^2 + \frac{dy}{dx})$ R-2.2-6<br>**c.** $3e^x \tan y\, dx + (1-e^x)\sec^2 y\, dy = 0$ R-2.3-7<br>**d.** $\sqrt{1+x^2+y^2+x^2y^2} + xy\frac{dy}{dx} = 0$ R-2.3-8<br>**e.** $\frac{dy}{dx} = e^{x+y} + x^2 e^{x^3+y}$ R-2.3-9<br>**f.** If $\frac{dy}{dx} = e^{x+y}$ and it is given for x = 1, y = 1; find y when x = -1 R-2.4-10<br>**g.** $x\sqrt{1+y^2}\, dx + y\sqrt{1+x^2}\, dy = 0$ R-2.4-Exercise 2(A)-3<br>**h.** $(e^y + 1)\cos x\, dx + e^y \sin x\, dy = 0$ R-2.4-Exercise 2(A)-6<br>**i.** $(x^2 - yx^2)dx + (y^2 + xy^2)dy = 0$ R-2.4-Exercise 2(A)-8 | |

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| | | **j.** $(xy^2+x)dx+(yx^2+y)dy=0$ R-2.4-Exercise 2(A)-9<br>**k.** $(1-x^2)(1-y)dx+xy(1+y)=0$ R-2.4-Exercise 2(A)-13<br>**l.** $\frac{dy}{dx}\tan y=\sin(x+y)+\sin(x-y)$ R-2.4-Exercise 2(A)-15 | |
| **1.11** | **Problems** **(**Transformation of some equations in the form in which variables are separable**)** | Transformation of some equations in the form in which variables are separable:<br>**a.** Standard form R-2.4-2.4<br>$\frac{dy}{dx}=f(ax+by+c)$ or $\frac{dy}{dx}=f(ax+by)$<br>Let $ax+by+c=v$ or $ax+by=v$<br>**b.** Working rule to solve an equation in the form in which variables are separable R-2.4<br><br>**Solve the following differential equations:**<br>**a.** $\frac{dy}{dx}=(4x+y+1)^2$ R-2.4-2.5-1<br>**b.** $\frac{dy}{dx}=\sin(x+y)+\cos(x+y)$ R-2.4-2.5-4<br>**c.** $\frac{dy}{dx}=\frac{4x+6y+5}{3y+2x+4}$ R-2.6-6<br>**d.** $(x+y+1)\frac{dy}{dx}=1$ R-2.7-Exercise 2(B)-5<br>**e.** $(2x+2y+3)dy-(x+y+1)dx=0$ R-2.7-Exercise 2(B)-10 | |
| **1.12** | **Problems** **(**Homogeneous differential equation**)** | Homogeneous differential equation:<br>**a.** Standard form R-2.7-2.6<br>$\frac{dy}{dx}=\frac{M(x,y)}{N(x,y)}$ where M and N have same degree or $\frac{dy}{dx}=f(\frac{y}{x})$<br>Let $\frac{y}{x}=v$ or $y=vx$<br>**b.** Working rule to solve homogeneous differential equation R-2.7-2.7<br><br>**Solve the following differential equations:**<br>**a.** $\left(x^3+3xy^2\right)dx+\left(y^3+3x^2y\right)dy=0$ R-2.8-2.8-1<br>**b.** $xdy-ydx=\sqrt{x^2+y^2}dx$ R-2.8-2.8-2<br>**c.** $\frac{dy}{dx}=\frac{y}{x}+\tan\frac{y}{x}$ R-2.8-2.8-3<br>**d.** $x\cos(\frac{y}{x})(ydx+xdy)=y\sin(\frac{y}{x})(xdy-ydx)$ R-2.9-4<br>**e.** $(4y+3x)dy+(y-2x)dx=0$ R-2.9-5<br>**f.** $x^2ydx-(x^3+y^3)dy=0$ R-2.11-Exercise 2(C)-10<br>**g.** $(x+y)dy-(x-y)dx=0$ R-2.11-Exercise 2(C)-11<br>**h.** $x\sin(\frac{y}{x})\frac{dy}{dx}=y\sin(\frac{y}{x})-x$ R-2.11-Exercise 2(C)-10-14 | |
| **1.13** | **Problems** **(**Equation Reducible to | Equation reducible to homogeneous form:<br>**a.** Standard form R-2.11-2.9 | |

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| | Homogeneous Form**)** | $\frac{dy}{dx}=\frac{ax+by+c}{a'x+b'y+c'}$ $as\frac{a}{a'}\neq\frac{b}{b'}$<br>Let $x=X+h$ and $y=Y+K$ where $X$ and $Y$ are new variables and h and k are constants<br><br>**b.** Working rule to solve equation reducible to homogeneous form R-2.11-2.9<br><br>**Solve the following differential equations:**<br>**a.** $\frac{dy}{dx}=\frac{x+2y-3}{2x+y-3}$ R-2.12-2.10-1<br>**b.** $\frac{dy}{dx}+\frac{x-y-2}{x-2y-3}=0$ R-2.12-2.10-2<br>**c.** $\frac{dy}{dx}=\frac{y-x-1}{y+x+5}$ R-2.15- Exercise 2(D)-2<br>**d.** $(3y-7x+7)dx+(7y-3x+3)dy=0$ R-2.15- Exercise 2(D)-10 | |
| **1.14** | **Problems (**Exact Differential Equation) | Exact Differential Equation:<br>**a.** Standard form R-2.16-2.12<br>$Mdx+Ndy=o\ if\ \frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$<br>**b.** Working rule to solve exact differential equation R-2.17-2.14<br><br>**Solve the following differential equations:**<br>**a.** $\left(1+e^{\frac{x}{y}}\right)dx+e^{\frac{x}{y}}(1-\frac{x}{y})dy=0$ R-2.18-4<br>**b.**<br>**c.** $\left\{y\left(1+\frac{1}{x}\right)+\cos y\right\}dx+(x+\log x-x\sin y)dy=0$ R-2.19-7<br>**d.** $y\sin 2x\,dx-(y^2+\cos^2 x)dy=0$ R-2.20-10(a) | |
| **1.15** | **Problems (Linear Differential Equation**) | **Linear Differential Equation**:<br>**a.** Standard form R-2.32-2.23<br>$\frac{dy}{dx}+Py=Q$<br>$\text{I.F.}=e^{\int P\,dx}$<br>**b.** Working rule to solve linear differential equation R-2.33-2.23<br><br>**Solve the following differential equations:**<br>**a.** $x\cos x(\frac{dy}{dx})+y(x\sin x+\cos x)=1$ R-2.33-2.24-1<br>**b.** $(x+2y^3)\frac{dy}{dx}=y$ R-2.35-5<br>**c.** $(1+y^2)dx=(\tan^{-1}y-x)dy$ R-2.35-6<br>**d.** $x(1-x^2)dy+(2x^2y-y-ax^3)dx=0$ R-2.37-8<br>**e.** $(x\log x)\frac{dy}{dx}+y=2\log x$ R-2.38-Exercise 2(K)-9 | |

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| | | **f.** $x^2(x^2-1)\frac{dy}{dx}+x(x^2+1)y=x^2-1$ R-2.38-Exercise(K)-13 | |
| **1.16** | **Problems (Equation Reducible to Linear Form)** | **Equation Reducible to Linear Form**:<br>**a.** Standard form R-2.38-2.25<br>$f'(y)\frac{dy}{dx}+Pf(y)=Q \text{ or } \frac{dy}{dx}+Py=Qy^n$<br>Let $f(y)=v$ or $y^{1-n}=v$<br>**b.** Working rule to solve equation reducible to linear form R-2.38-2.25<br>**c.** Bernoulli's equation R-2.39-2.25A<br>$f'(y)\frac{dy}{dx}+Pf(y)=Q \text{ or } \frac{dy}{dx}+Py=Qy^n$<br>Let $f(y)=v$ or $y^{1-n}=v$<br>**d.** Working rule to solve Bernoulli's equation R-2.39-2.25A<br><br>**Solve the following differential equations:**<br>**a.** $\frac{dy}{dx}+x\sin 2y=x^3\cos^2 y$ R-2.39-2.26-1<br>**b.** $\frac{dy}{dx}=e^{x-y}(e^x-e^y)$ R-2.40-2<br>**c.** $x\frac{dy}{dx}+y\log y=xye^x$ R-2.40-4<br>**d.** $\frac{dy}{dx}-\frac{\tan y}{1+x}=(1+x)e^x\sec y$ R-2.42-Exercise 2(L)-4<br>**e.** $x\frac{dy}{dx}+y=y^2\log x$ R-2.43-2.27-1<br>**f.** $(x^3y^2+xy)dx=dy$ R-2.43-2.27-3(a) | |
| **1.11** | **Sample Questions** | Some of sample questions are:<br>**a.** Evaluate the general solution of the following differential equations (any two including i):<br>**i.** $\sqrt{1+x^2+u^2+x^2u^2}+xu\frac{du}{dx}=0$<br>**ii.** $(2u+2v+3)dv-(u+v+1)du=0$<br>**iii.** $x\sin(\frac{u}{x})\frac{du}{dx}=u\sin(\frac{u}{x})-x$<br>**iv.** $x(1-x^2)du+(2x^2u-u-ax^3)dx=0$<br>**v.** $\frac{dv}{dx}=e^{x-v}(e^x-e^v)$<br><br>**Marks** 12 **CO** 2 **PO** 2 **BL** C3, C4, C5 **KP** WK2<br><br>**b.** Compute the general solution of any three of the following differential equations, including i: | Remember (C1), Understand (C2), Apply (C3), Analyze (C4), Evaluate (C5), Create (C6) |

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i. $(e^y+1)\cos x\,dx + e^y \sin x\,dy = 0$<br>
ii. $x\cos(\frac{y}{x})(ydx+xdy) = y\sin(\frac{y}{x})(xdy-ydx)$<br>
iii. $(x+2y^3)\frac{dy}{dx} = y$<br>
iv. $x\frac{dy}{dx} + y = y^2 \log x$
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c. Assess the general solution of the Bernoulli's differential equation $x\frac{dy}{dx} + y\log y = xye^x$.
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d. Deliberate the working steps to solve linear differential equation. Calculate the general solution of the differential equation $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$.
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<td>1.12</td>
<td>Others</td>
<td>For practice<br>R-1.9- Exercise-1(A)<br>AY-16- Exercise-1</td>
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